

Extended Echo State Network Models for Analyzing Long-Term Dependencies in Real-Time Tasks

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Goal

The main goal of my work is to find an alternative way to initialize reservoirs to further improve performance of Echo State Networks in time series forecasting tasks.



Problem: Time Series Forecasting

Time series - a sequential set of data points measured over successive times. Examples include: temperature reading, production output, stock market data. It can be modelled as a stochastic process. Usually 4 main components are separated from the data: Trend, Cyclical, Seasonal and Irregular components [1].

Time series forecasting - prediction of future values of the time series, based on its history and possibly the external data. It has important applications in various fields: finance, engineering, medicine [1].

Examples of times series



Fig 1. Weekly BP/USD exchange rate series (1980-1993) [1]

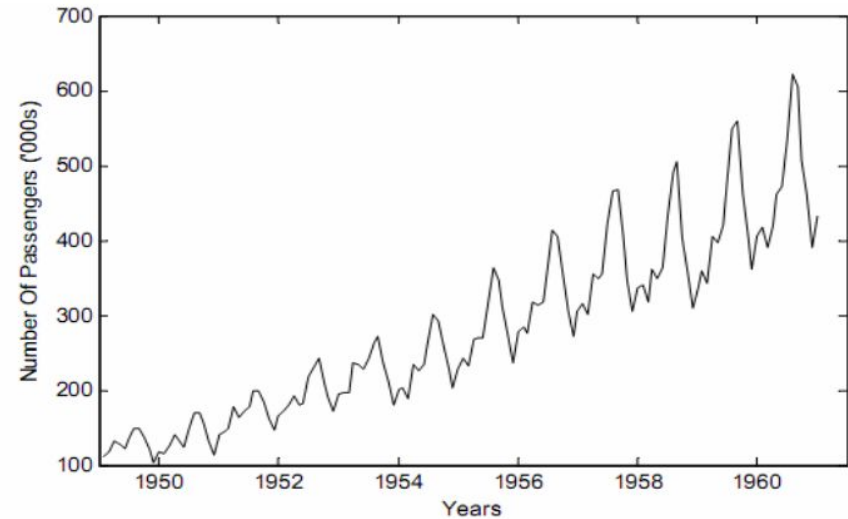


Fig 2. Monthly international airline passenger series (Jan. 1949-Dec. 1960) [1]



Time Series Forecasting: Existing Solutions

We can divide existing approaches in the following categories [2]:

- Statistical models
- Machine Learning Models
- Deep Learning
- Hybrid Deep Learning Methods



Traditional approaches

Statistical models

Prior to machine learning, statistical models were primarily used for time series forecasting [2].

Methods like exponential smoothing [3] or ARIMA (Autoregressive Integrated Moving Average) [4] were a standard

Machine learning

Machine learning methods excel in learning nonlinear patterns that are often present in time series [2].

Many traditional ML approaches were applied to the forecasting problem: classification and regression trees [5], support vector regression [6], XGBoost [7], any many more.



Deep learning

Advent of Multi-layer Perceptrons [8] resulted in many attempts in applying them to time series forecasting problems. They, however, suffered from multiple issues in this domain [2]:

- Limited in learning temporal dependencies
- Vanishing gradients
- Lack of data and computing power

With time, many models were found useful in the task of forecasting: CNNs, RNNs, transformers. Nowadays, hybrid methods and their adaptations are often used.

Recurrent Neural Networks



Development of Recurrent Neural Networks [9] opened up new possibilities for processing of time series data. Its structure of the hidden layers gives us a natural way to analyze sequential data of arbitrary length.

They still faced multiple issues: vanishing and exploding gradients (in BPTT) and difficulties with parallel processing. Multiple variants were proposed to overcome these issues: LSTM, GRU, uRNN. Modern ones include: SSM, MAMBA, linear RNN [2].

The main equations describing RNNs are as follows [10]:

$$h_t = \phi_h(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

$$o_t = \phi_o(W_o h_t + b_o)$$

h_t - hidden state at timestep t

x_t - input at timestep t

W_{xh}, W_{hh}, W_o - weight matrices

b_h, b_o - bias vectors

ϕ_h, ϕ_o - nonlinear activation functions

o_t - output at timestep t

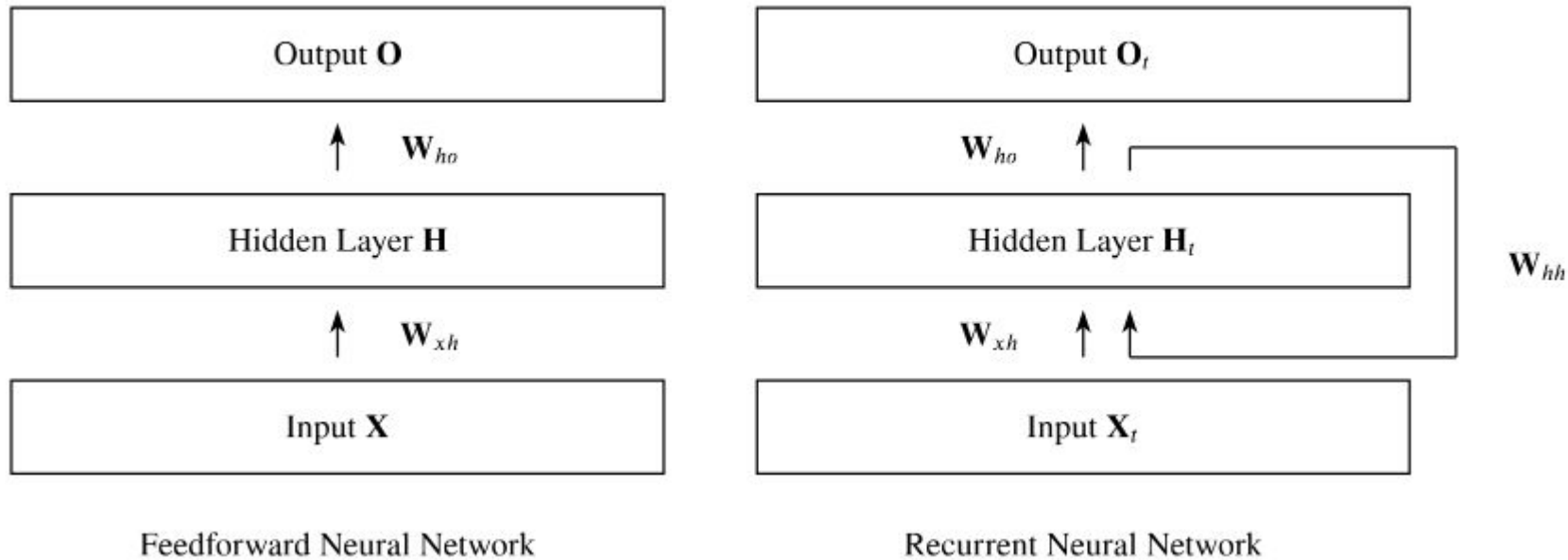


Fig 3. Visualisation of differences between MLPs and Recurrent Neural Networks [10]



Transformers

Transformers [11] emerged as a dominant method for sequence-based problems, like NLP. This also prompted research in their applications to time series forecasting.

They address two significant issues that RNNs faces: long-term dependencies and parallel processing capabilities. This is achieved through the attention mechanism. However, new problems arise when applying transformers in time series forecasting tasks:

- Efficient challenges
- Inefficiencies of expanding input windows length [12]
- Points in time series are locally similar and carry limited information [13]
- Overfitting to noise [2]



Variants of transformers

Multiple new variants of transformer architectures were proposed to address time series forecasting tasks. In general, following approaches were suggested:

- Patching mechanisms [14]
- Cross-dimension [15]
- Exogenous variables [26]



Echo State Networks

Echo State Networks are variants of RNNs. They present an alternative to gradient-based learning of recurrent weights. Two major changes are introduced:

- Instead of traditional recurrent hidden layer, a reservoir is introduced. It can be represented as a large, random, sparse matrix. The reservoir is fixed - it is not trained during training. It completely avoids BPTT.
- Only the readout is trained. The weights are usually trained with ridge regression.

Echo State Networks show great performance in time series forecasting and dynamical systems modelling. They are also extremely efficient compares to other methods [16].

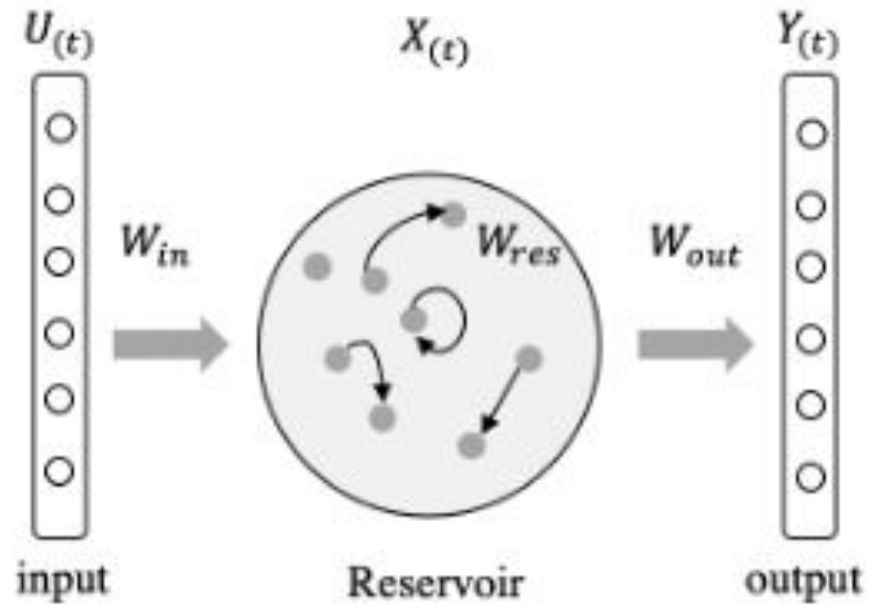


Fig 4. Basic visual representation of Echo State Networks [16]



Echo State Networks

The ESN can be characterized by the following equations [17]:

$$\bar{h}_t = \phi_h(W_{xh}x_t + W_{hh}h_{t-1})$$

$$h_t = \alpha \bar{h}_t + (1 - \alpha)h_{t-1}$$

$$o_t = W_o h_t + b_o$$

h_t - hidden state at timestep t

x_t - input at timestep t

W_{xh}, W_o - weight matrices

W_{hh} - reservoir weights

b_o - bias vector

ϕ_h, ϕ_o - nonlinear activation functions

o_t - output at timestep t

α - leaky rate



Problem: How to initialize Echo State Networks?

The reservoir weights cannot just be any random matrix. It is known that the reservoir initialization strategy has a crucial effect on the final capability of the model. The most important is **echo state property**.

Echo state property - for a long enough input, the hidden state of the system should not depend on the initial state. It is usually achieved by controlling the spectral radius of the reservoir.

Traditionally, ESN reservoir is initialized with a random sparsity mask ensuring little connectivity (e.g. 10 connections per node), nonzero elements are sampled from uniform or Gaussian distribution, and spectral radius is set close to 1 for tasks requiring long term memory [17].



My Approach: Motivation

1. The performance of the Echo State Networks can be approximately inferred from by eigenvalue analysis [18].
2. Adapting timescales of the ESN's reservoir can lead to improved performance. Timescales are directly associated with eigenvalues [19].
3. Edge of Stability Networks show us that eigenvalues close to a unit circle in the complex plane provide the system with better memory. It is achieved by altering the hidden state update equation with a linear projection [20].

$$h_t = \beta \phi_h(W_{xh}x_t + W_{hh}h_{t-1}) + (1 - \beta)Oh_{t-1}$$

β - proximity parameter

O - random orthogonal matrix



My Approach: Motivation

Different tasks can require their own sets of timescales and eigenvalues.

What if we could directly control the distribution of eigenvalues in the reservoir instead of relying in hyperparameter tuning or altering system's equations?



My Approach: Isospectral Echo State Networks

The main idea of my approach is to first sample the eigenvalues and then construct the reservoir based on them. The process looks like this:

1. Choose a distribution to sample from. It should have some tunable hyperparameters p .
2. Sample eigenvalues.
3. Construct a reservoir with given eigenvalues and a random sparsity mask. It can be achieved via Stiefel manifold optimization and similarity transformation or Hungarian algorithm and Schur decomposition.
4. Train the model and validate its performance.
 - a. If it's good enough, finish.
 - b. If not, adapt the hyperparameters p and go back to step 2. I use Bayesian optimization for this step.



Spectrum Comparison

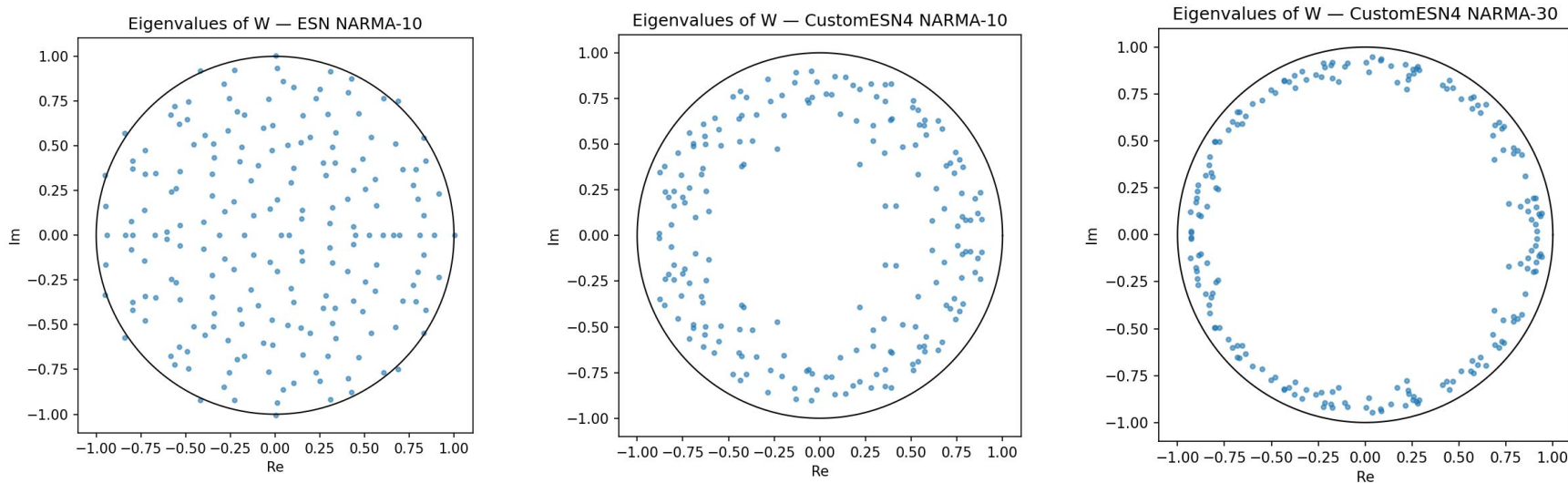


Fig 5. Eigenvalues of: a) an example standard ESN b) two examples of different isospectral ESNs



Reservoir Construction Approaches

Stiefel manifold optimization + similarity transformation

1. Arrange sampled eigenvalues in a diagonal matrix D
2. Optimize an orthogonal matrix Q on a Stiefel manifold minimizing the sum of squared magnitudes of $W = QDQ^*$, so that it matches the sparsity mask
3. Apply a hard threshold to the W matrix to zero out small values (so that the matrix is actually sparse)

Hungarian algorithm + Schur decomposition

1. Precompute T (upper triangular) and Z (unitary) templates so that $W = ZTZ^* \odot M$ (sparsity mask) has a set of given placeholder eigenvalues via iterative Schur decomposition + Hungarian matching
2. Replace the diagonal of T with actual eigenvalues via Hungarian matching, reconstruct $W = ZTZ^* \odot M$, and repeat for n iterations.



Benefits

Isospectral Echo State Networks provide us following benefits:

- Full control of the spectrum
- Task specific tuning of the model's memory patterns
- Greater adjustability of reservoir weights compared to traditional ESNs.
Sampling distribution of the eigenvalues can be freely chosen according to the needs.
- Preserves efficiency and rich dynamics of traditional ESNs



Applications

Echo State Networks can be applied to various types of time series:

- Chaotic time series prediction [21]
- Speech recognition and audio processing [22]
- Robotics and control systems [23]
- Biomedical signal processing [24]
- Industrial systems modelling [25]



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Thank you!